

AWS Notes: SSH, Security Groups and Instance Types

Connecting to an EC2 instance with PowerShell, PuTTY and EC2 Instance Connect

1. SSH Overview

SSH (Secure Shell) lets you control a remote Linux machine from your own computer by typing commands in a terminal. It runs on **port 22** and uses a **key pair**: AWS puts the **public key** on the instance and you keep the **private key** file (.pem). Whoever holds the private key can log in, so protect it.

Where you work	Best way to connect
Windows 10 / 11	PowerShell or Command Prompt (OpenSSH is built in), or PuTTY
Older Windows	PuTTY (needs a .ppk key made from the .pem)
Mac / Linux	Built-in Terminal with the ssh command
Any OS, just a browser	EC2 Instance Connect (no key file needed)

Default user names: `ec2-user` on Amazon Linux, `ubuntu` on Ubuntu.

2. Security Groups

A security group is a **virtual firewall around an instance**. It controls what traffic may come in (inbound) and go out (outbound).

- **Allow rules only.** You cannot write deny rules. Anything not allowed is blocked.
- **Stateful:** if inbound traffic is allowed, the reply goes out automatically.
- A new security group has **no inbound rules** (everything blocked in) and **allows all outbound**.
- Rules use a protocol, a **port** and a **source** (an IP range like My IP, 0.0.0.0/0 for anywhere, or another security group).
- One instance can have several security groups, and one group can be used by many instances. Groups belong to a Region and VPC.

Port	Protocol	Used for
22	SSH / SFTP	Remote login to Linux, secure file transfer
80	HTTP	Websites (unencrypted)
443	HTTPS	Websites (encrypted)
3389	RDP	Remote desktop to Windows
21	FTP	File transfer (unencrypted)

Timeout vs refused (a handy debugging trick): a connection that **times out** is almost always a security group problem (port not allowed). A connection that is **refused** means traffic reached the instance but nothing is listening on that port.

3. EC2 Instance Types

Names follow a pattern, e.g. `m5.2xlarge`: **m** = family, **5** = generation, **2xlarge** = size (bigger size = more vCPU and memory). A `t3.micro`, as in your account, is a small general purpose burstable type.

Family	Optimised for	Examples
General purpose	Balanced CPU, memory, network. Web servers, small apps.	t3, m5
Compute optimised	CPU-heavy: batch jobs, gaming, media encoding.	c5, c6i
Memory optimised	Large in-memory data: databases, caches.	r5, x1
Storage optimised	High disk speed: data warehouses.	i3, d2
Accelerated	GPU / machine learning.	p3, g4

The **t-family is burstable**: it collects CPU credits when idle and spends them under load, which suits learning.

4. Hands-on: Control the Machine from PowerShell

- 1 Open PowerShell and go to the folder with your key: `cd C:\Users\User\Downloads\aws-course`.
- 2 Make sure the instance is **running** and its security group allows **SSH (22) from My IP**.
- 3 Connect (copy the public IPv4 from the console). Type **yes** the first time to trust the host:

```
ssh -i .\EC2Tutorial.pem ec2-user@<public-ip>
```

When the prompt changes to `[ec2-user@ip-172-31-... ~]$`, every command now runs on the instance, not your PC. Try:

```
whoami          # ec2-user
hostname        # the instance's private DNS name
cat /etc/os-release # which Linux this is
pwd ; ls        # where you are, what is here
sudo yum install -y httpd # install a web server
sudo systemctl start httpd
echo "<h1>Hello from EC2</h1>" | sudo tee /var/www/html/index.html
exit            # back to your own PC
```

Then add an inbound rule **HTTP, port 80, source Anywhere** to the security group and open `http://<public-ip>` in a browser (http, not https). It shows your page. This proves a security group rule decides who can reach the server. Remove the rule when finished.

5. Hands-on: PuTTY

- 1 Open **PuTTYgen** > **Load** your .pem (set the file filter to All Files) > **Save private key** to make a .ppk.
- 2 Open **PuTTY**. Under Session, Host Name: `ec2-user@<public-ip>`, Port 22, type SSH.
- 3 Go to **Connection > SSH > Auth > Credentials**. Put the .ppk in **Private key file for authentication**. **Leave the Certificate box empty**.
- 4 Back on Session, name it, click **Save**, then **Open**. Accept the host key warning.

6. Hands-on: EC2 Instance Connect

- 1 EC2 > Instances > select the running instance > **Connect**.
- 2 Open the **EC2 Instance Connect** tab, check the user name, click **Connect**. A browser terminal opens.

It needs no key file because AWS pushes a temporary key (about 60 seconds) for you. It still needs port 22 reachable and a public IP, and works on Amazon Linux and Ubuntu (not Windows instances).

7. Troubleshooting

Symptom	Cause and fix
PuTTY: Unable to use certificate file ... is not a certificate	The .ppk is in the Certificate box. Clear it; the key belongs only under Private key file.
PuTTY: No supported authentication methods available	The server rejected the key, or it is not the key pair used at launch. Check Key pair assigned at launch on the instance Details tab.
PowerShell: UNPROTECTED PRIVATE KEY FILE / bad permissions	Other accounts can read the .pem, so ssh ignores it. Fix with the icac ls commands below.
Permission denied (publickey)	Wrong key or wrong user name (ec2-user vs ubuntu).
Connection timed out	Security group does not allow port 22 from your current IP, or the public IP changed after a stop and start.
Connection refused	Reached the instance but nothing listens on that port (service not started).

Fix key permissions on Windows (run in the key's folder; use the account name shown in the error for the second line):

```
icac ls .\EC2Tutorial.pem /inheritance:r
icac ls .\EC2Tutorial.pem /remove "ACCOUNT-NAME-FROM-THE-ERROR"
icac ls .\EC2Tutorial.pem /grant:r "${env:USERNAME}:R"
```

8. Good Habits

- Allow SSH only from **My IP**, not 0.0.0.0/0.
- Never share or upload .pem or .ppk files. Add * .pem and * .ppk to a .gitignore file in any repo.
- A key pair cannot be changed after launch. Keep the file safe or use Instance Connect.
- Public IPs change when an instance is stopped and started. Stop instances when not studying to save money.

9. Check Yourself

Question	Answer
Which port does SSH use?	22
Which key stays on your computer?	The private key (.pem or .ppk).
Are security groups allow-only or allow and deny?	Allow only. Everything else is blocked.
Connection times out. First thing to check?	The security group rule for that port and your IP.
What does m5.2xlarge mean?	Family m, generation 5, size 2xlarge.
Which method needs no key file?	EC2 Instance Connect.